

Linear Regression Using Top 6 Principal Components

1. Load Libraries

```
library(ggplot2)
library(dplyr)
library(car)      # for VIF
library(caret)    # for RMSE and data splitting
```

2. Load and Filter the Data

```
# Load cleaned Boston dataset
data <- read.csv("../data/boston_cleaned.csv")

# Check structure
str(data)
```

```
## 'data.frame':  506 obs. of  13 variables:
## $ CRIM      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ ZN        : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ INDUS     : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ CHAS      : int   0 0 0 0 0 0 0 0 0 0 ...
## $ NOX       : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ RM        : num  6.58 6.42 7.18 7 7.15 ...
## $ AGE       : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ DIS       : num  4.09 4.97 4.97 6.06 6.06 ...
## $ RAD       : int   1 2 2 3 3 3 5 5 5 5 ...
## $ TAX       : int  296 242 242 222 222 222 311 311 311 311 ...
## $ PTRATIO   : num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ LSTAT     : num  4.98 9.14 4.03 2.94 5.33 ...
## $ MEDV      : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

3. Principal Component Analysis (PCA)

```
# Run PCA on standardized predictors
pr.out <- prcomp(data %>% select(-MEDV), scale. = TRUE)

# Extract top 6 principal components
pc_data <- as.data.frame(pr.out$x[, 1:5])
pc_data$MEDV <- data$MEDV
```

4. Split Data (80/20)

```
# Set seed for reproducibility
set.seed(2641)

# Split into training (80%) and testing (20%)
splitIndex <- createDataPartition(pc_data$MEDV, p = 0.8, list = FALSE)
train <- pc_data[splitIndex, ]
test <- pc_data[-splitIndex, ]
```

5. Fit Linear Regression Model

```
# Fit model on training data
PCReg <- lm(MEDV ~ ., data = train)
summary(PCReg)
```

```
##
## Call:
## lm(formula = MEDV ~ ., data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.431  -2.742  -0.767   1.934  33.432
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.4955     0.2664  84.430 < 2e-16 ***
## PC1          -2.2606     0.1109 -20.392 < 2e-16 ***
## PC2           2.7629     0.2194  12.594 < 2e-16 ***
## PC3           3.1605     0.2436  12.972 < 2e-16 ***
## PC4           1.7431     0.2786   6.256 1.01e-09 ***
## PC5          -1.4640     0.2982  -4.910 1.33e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.369 on 401 degrees of freedom
## Multiple R-squared:  0.6642, Adjusted R-squared:  0.66
## F-statistic: 158.7 on 5 and 401 DF, p-value: < 2.2e-16
```

6. Predict & Evaluate on Test Set

```
# Predict on test set
predictions <- predict(PCReg, newdata = test)

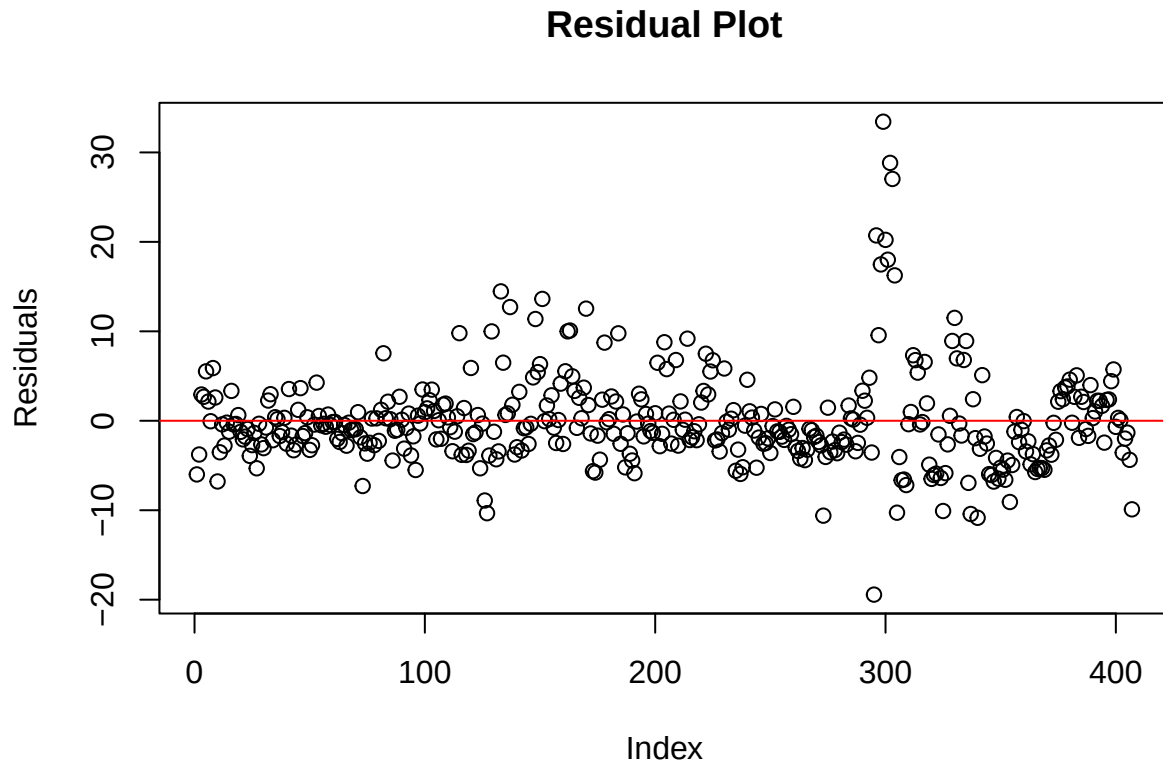
# Calculate SSE and SST
SSE <- sum((test$MEDV - predictions)^2)
ybar <- mean(train$MEDV)
SST <- sum((test$MEDV - ybar)^2)
```

```
# Out-of-sample R-squared
R2_out <- 1 - SSE/SST
cat("Out-of-sample R2:", round(R2_out, 4))
```

```
## Out-of-sample R2: 0.7969
```

7. Residual Plot

```
# Residual plot
plot(PCReg$residuals, main = "Residual Plot", ylab = "Residuals", xlab = "Index")
abline(h = 0, col = "red")
```



8. Model Validation

a. T-tests and F-test (from `summary()`)

```
summary(PCReg) # includes t-values and F-statistic
```

```
##
## Call:
```

```
## lm(formula = MEDV ~ ., data = train)
##
## Residuals:
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## -19.431  -2.742  -0.767   1.934  33.432
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```

b. VIF (Variance Inflation Factor)

```
vif(PCReg) # check multicollinearity
```

```
##      PC1      PC2      PC3      PC4      PC5
## 1.001624 1.001862 1.002673 1.000963 1.001675
```

Summary

- The model uses **PC1–PC5** which together explain ~85% of variance in the predictors.
- The **out-of-sample R^2** is 0.7969, showing how well the model generalizes.
- **No multicollinearity** is expected since PCs are orthogonal (VIF ~ 1).
- Model significance assessed via **t-tests** and **F-test**.